

K B Chandrashekar

+918951088256 • chandrabala171@gmail.com • Dindigul, India 624301

Objective

Results-driven **Electronics and Communication Engineering graduate** (expected 2025) from Assam University, with hands-on experience in **VLSI design, FPGA development, and IoT solutions**. Proven leadership as **Class Representative** and technical expertise in optimizing systems through machine learning and hardware design. Seeking to leverage my **technical skills, problem-solving ability, and team management experience** to contribute to innovative engineering projects.

Technical Skills

- C
- MATLAB
- Python
- Cadence Virtuoso
- Vivado
- Proteus
- Tinkercad
- VLSI
- FPGA
- PCB Design
- IoT
- Arduino
- Raspberry Pi
- Sensor Networks
- SPICE
- LTspice
- Machine Learning
- Microsoft Office
- AutoCAD
- Linux
- Computer Networking

Experience

06/2023 - 06/2025

Class Representative, **Assam University**, Silchar, IN

- Distributed information regarding upcoming events to members of the class.
- Built surveys to collect information from students at different levels and in different majors.
- Attended monthly meetings to voice concerns, opinions and ideas.
- Advocated for student rights in administrative forums.
- Maintained accurate records of student attendance and performance data.
- Maintained an open dialogue with classmates about their concerns and issues faced by them.

11/2024 - 11/2024

Intern, **Assam University**, Silchar, IN

- Completed intensive hands-on training in **ASIC design flow** (180nm technology node).
- Designed and simulated **CMOS logic gates** (INVERTER, NAND, NOR) with 95% accuracy in SPICE verification.

06/2024 - 07/2024

FPGA Design Intern, **NIT Silchar**

- Engineered high-speed data acquisition systems utilizing Xilinx FPGAs.
- Optimized Verilog code, achieving a 30% reduction in latency.

Expected in 06/2025

Education

Bachelor Of Technology, Electronics and Communication Engineering
Assam University, Silchar

Pre-University Course (PUC)
Sarvajna PU College, Gulbarga, Karnataka

Websites, Portfolios, Profiles

- [linkedin.com/in/k-b-chandrashekaran-642a1a357](https://www.linkedin.com/in/k-b-chandrashekaran-642a1a357)

Certifications

- VLSI Design, Assam university, 2024
- FPGA Design, NIT Silchar, 2024
- MATLAB for Engineers, NIT Silchar, 2024
- C Programming, Assam university, 2022

Training

- VLSI Design Workshop, Entuple Technologies, 2023
- IoT & Embedded Systems, NIT Silchar, 2022
- Cadence Virtuoso Tool Training, Assam University, 2023

Languages

English, Tamil, Kannada, Telugu, Assamese, Malayalam, Bengali

Projects

- Op-Amp Aspect Ratio Optimization Using MATLAB GUI, developed a GUI-based tool to automate transistor sizing, reducing manual calculation time by 40%, and integrated ML algorithms to predict optimal parameters, improving design accuracy by 25% Verified results using Cadence Virtuoso for industrial-standard validation
- IoT-Based Smart Agriculture System, Designed a soil monitoring system using moisture/temperature sensors and ESP8266. Implemented real-time data logging via ThingSpeak, enabling remote farm management
- FPGA-Based Digital Signal Processor: Built a noise-filtering DSP using Verilog on Xilinx Vivado, achieving 90% noise reduction
- Designed a stationary monitoring station tracking patient vitals (heart rate, SpO₂, temperature) with less than 5% error margin versus clinical devices, transmitting real-time data to the cloud (ThingSpeak, Blynk) via secure APIs for remote healthcare access

Leadership And Extracurriculars

- Class Representative, 2022-2024, led peer mentoring initiatives and event planning
- Cooking enthusiast, experiment with fusion recipes, which showcase creativity and precision
- Automotive Enthusiast: Passionate about motorcycle mechanics and performance tuning. Enjoy applying engineering principles to diagnose systems and optimize vehicle performance through data analysis and hands-on modifications.

Accomplishments

- Led a six-member motorcycle expedition from Silchar to Tawang's Bum La Pass (Indo-China border, 750 km), coordinating route planning, safety protocols, and mechanical support for

high-altitude terrain (15,200 ft).